

Coupled Einstein-Spinor Dynamics in Curved Signature $(4,4)$

A standalone numerical and analytic provenance report for one real rank-16 spinor bundle

Reproducible exact-real implementation

September 2026

Contents

1	Base geometry and the spinor bundle	3
2	Canonical spin connection used by the model	4
3	Exact real gamma matrices and invariant adjoint	5
4	Classical-field assumption	6
5	Action without a scalar quintessence field or cosmological constant	6
6	Stress tensor and homogeneous reduction	7
7	Signature $(4,4)$ Einstein equations	7
8	Chosen nonlinear spinor potential	9
9	Complete monomial connection map	9
10	Acceleration transition	11
11	Numerical initial-value problem	11
12	Canonical numerical results	12
13	Independent verification	13
14	What may be connected to dark matter	14

15 What may be connected to dark energy	15
16 Other possible connections and excluded claims	15
17 Complete Windows commands	16
18 Complete Git Bash or WSL commands	18
19 Files and ownership boundaries	20
20 Conclusion	21

Abstract

This document defines and approximately solves a coupled Einstein-spinor model on an explicit curved 8-dimensional pseudo-Riemannian manifold of signature $(4,4)$. The matter field is one commuting classical real 16-component $\text{Spin}(4,4)$ spinor, locally decomposed into the two inequivalent real 8-component half-spin modules. There is no quintessence scalar and no cosmological-constant term. A nonlinear spinor potential supplies both a rapidly diluting dust-like component and a slowly diluting negative-pressure component.

The metric, vielbein, Christoffel symbols, and canonical Levi-Civita spin connection are derived explicitly. A symmetric nondegenerate split bilinear $\mathbb{C}$ defines the local Spin-invariant condensate $\mathbb{S} = \bar{\psi} \mathbb{C} \psi$. The action, Dirac equation, stress tensor, and 8-dimensional Einstein equations are stated with all conventions. The homogeneous system is integrated with the pinned pure-Rust SUNDIALS CVODE implementation. Independent checks recompute every emitted thermodynamic quantity, both Einstein equations, all 18 differential equations by five-point finite differences, the analytic condensate and density laws, the acceleration transition, source hashes, and byte-identical replay.

This is a split-signature mathematical model, not standard one-time cosmology. Its constant-time slices have signature $(4,3)$. Terms are called "dust-like" and "negative-pressure" rather than dark matter and dark energy unless a comparison is explicitly qualified. The calculation does not prove causal stability, quantum consistency, compactification, observational viability, or a microscopic particle interpretation.

1 Base geometry and the spinor bundle

Let (M, g) be an oriented, time-oriented, spin 8-manifold with metric signature $(4,4)$. Use coordinates

$$(x_0, x_1, x_2, x_3, t, y_1, y_2, y_3)$$

and constant tangent metric

$$\eta_{ab} = \text{diag}(1, 1, 1, 1, -1, -1, -1, -1).$$

The selected curved metric is

$$ds^2 = a(t)^2 \left[(dx^0)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2 - (dy^1)^2 - (dy^2)^2 - (dy^3)^2 \right] - dt^2.$$

The diagonal vielbein and inverse are

$$e_\mu^a = \text{diag}(a, a, a, a, 1, a, a, a),$$

$$e_a^\mu = \text{diag}(a^{-1}, a^{-1}, a^{-1}, a^{-1}, 1, a^{-1}, a^{-1}, a^{-1}),$$

and satisfy

$$g_{\mu\nu} = e_\mu^a \eta_{ab} e_\nu^b.$$

The principal pseudo-orthonormal frame bundle is lifted to a principal $\mathbf{Spin}(4,4)$ bundle. The associated real rank-16 spinor bundle is

$$\mathcal{S} = P_{\mathbf{Spin}(4,4)}(M) \times_{\rho} \Delta_{\mathbb{R}},$$

with

$$\Delta_{\mathbb{R}} = \Delta_+ \oplus \Delta_-, \quad \dim \Delta_+ = \dim \Delta_- = 8.$$

A field is

$$\psi(t, x) \in \Gamma(\mathcal{S}), \quad \psi = \psi_+ \oplus \psi_-.$$

The spin connection is even and preserves the two half-spin subbundles. The Dirac operator includes odd Clifford multiplication and can couple the two chiral components.

2 Canonical spin connection used by the model

The Levi-Civita Christoffel symbols are

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_{\mu} g_{\nu\sigma} + \partial_{\nu} g_{\mu\sigma} - \partial_{\sigma} g_{\mu\nu}).$$

The vielbein postulate

$$\partial_{\mu} e_{\nu}{}^a - \Gamma^{\rho}_{\mu\nu} e_{\rho}{}^a + \omega_{\mu}{}^a{}_b e_{\nu}{}^b = 0$$

has the explicit solution

$$\omega_{\mu}{}^a{}_b = e_b{}^{\nu} (\Gamma^{\rho}_{\mu\nu} e_{\rho}{}^a - \partial_{\mu} e_{\nu}{}^a).$$

With $\mathbf{H}=\mathbf{adot}/\mathbf{a}$ and transverse indices $\mathbf{I}=\{0,1,2,3,5,6,7\}$, the nonzero components are

$$\Gamma^4_{ii} = \eta_{ii} a \dot{a}, \quad \Gamma^i_{4i} = \Gamma^i_{i4} = H,$$

$$\omega_{ii4} = \eta_{ii} \dot{a}, \quad \omega_{i4i} = -\eta_{ii} \dot{a}.$$

There are 21 nonzero Christoffel entries and 14 nonzero lowered spin-connection entries.

The spinor derivative is

$$D_{\mu} \psi = \partial_{\mu} \psi + \frac{1}{8} \omega_{\mu ab} [\gamma^a, \gamma^b] \psi,$$

where all ordered values of $\mathbf{a}, \mathbf{b}$ are summed. If only $\mathbf{a} < \mathbf{b}$ is summed, the equivalent coefficient is $1/4$.

For a homogeneous spinor,

$$\gamma^\mu D_\mu \psi = \gamma^4 \left(\partial_t + \frac{7}{2} H \right) \psi.$$

This $7H/2$ term is computed exactly in independent Python and Wolfram implementations.

3 Exact real gamma matrices and invariant adjoint

Use

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For $k=1, 2, 3, 4$, define

$$\gamma_k^+ = G^{\otimes(k-1)} \otimes P \otimes I_2^{\otimes(4-k)},$$

$$\gamma_k^- = G^{\otimes(k-1)} \otimes N \otimes I_2^{\otimes(4-k)}.$$

They are real integral 16 by 16 matrices satisfying

$$\{\gamma^a, \gamma^b\} = 2\eta^{ab} I_{16}.$$

The evolution-time matrix is the first negative generator,

$$\gamma^4 = \gamma_1^-, \quad (\gamma^4)^2 = -I_{16}.$$

Define

$$C = \gamma_1^+ \gamma_2^+ \gamma_3^+ \gamma_4^+.$$

Then

$$C^\mathsf{T} = C, \quad C^2 = I, \quad (\gamma^a)^\mathsf{T} C = -C \gamma^a.$$

The eigenvalue signature of $\mathbf{C}$ is $(8, 8)$. The real adjoint and condensate are

$$\bar{\psi} = \psi^\mathsf{T} C, \quad S = \bar{\psi} \psi.$$

This bilinear is locally $\mathbf{Spin}(4, 4)$ invariant and indefinite.

4 Classical-field assumption

The field in this numerical model is a commuting classical real spinor. This assumption is essential. For a single Grassmann-odd spinor, anticommuting components contracted with the symmetric matrix $\mathbf{C}$ would give

$$\psi^\top C \psi = 0.$$

A nonzero nonlinear potential $V(S)$ would then require independent dual fields, multiple fermionic species, or another bilinear. This study instead uses a commuting effective classical order parameter, analogous in modeling role to a classical condensate but transforming in the real spin bundle. Nothing here quantizes the field or identifies a fundamental fermion.

5 Action without a scalar quintessence field or cosmological constant

Choose the total action

$$I[g, \psi] = \int_M d^8x \sqrt{|g|} \left[\frac{1}{2\kappa_8} R + \mathcal{L}_\psi \right],$$

with

$$\mathcal{L}_\psi = \frac{1}{2} \left(\bar{\psi} \gamma^\mu D_\mu \psi - (D_\mu \bar{\psi}) \gamma^\mu \psi \right) - V(S).$$

No cosmological constant and no scalar field occur. The potential depends only on the invariant spinor condensate.

Varying the independent classical spinor variables gives

$$\boxed{(\gamma^\mu D_\mu - V_S) \psi = 0},$$

where

$$V_S = \frac{dV}{dS}.$$

For the homogeneous frame,

$$\gamma^4 \left(\dot{\psi} + \frac{7}{2} H \psi \right) - V_S \psi = 0.$$

Multiplying by $-(\gamma^4)$ gives the implemented real first-order equation

$$\dot{\psi} = -\frac{7}{2}H\psi - V_S\gamma^4\psi.$$

The sign is fixed by $(\gamma^4)^2 = -I$.

6 Stress tensor and homogeneous reduction

The symmetric spinor stress tensor is

$$T_{\mu\nu} = -\frac{1}{4} \left[\bar{\psi}\gamma_\mu D_\nu\psi + \bar{\psi}\gamma_\nu D_\mu\psi - (D_\mu\bar{\psi})\gamma_\nu\psi - (D_\nu\bar{\psi})\gamma_\mu\psi \right] + g_{\mu\nu}\mathcal{L}_\psi.$$

This overall sign follows from $T_{\mu\nu} = -(2/\sqrt{|g|}) \delta I_m / \delta g^{\mu\nu}$ together with the action and Einstein-equation convention used here. With the opposite definition of the metric variation, both the definition of T and the Einstein equation would have to change together.

For the homogeneous on-shell ansatz it takes the isotropic transverse form

$$T_{tt} = \rho, \quad T_{AB} = p g_{AB},$$

with

$$\rho = V(S),$$

$$p = SV_S - V.$$

The γ^4 term is skew-adjoint for $\mathbb{C}$, so it drops out of the condensate derivative:

$$\dot{S} = \dot{\psi}^\top C\psi + \psi^\top C\dot{\psi} = -7HS.$$

Consequently

$$S = S_0 a^{-7}.$$

This is also the continuity equation consequence

$$\dot{\rho} + 7H(\rho + p) = 0.$$

7 Signature (4,4) Einstein equations

Use

$$G_{\mu\nu} = \kappa_8 T_{\mu\nu}.$$

For the flat transverse metric with seven scaled directions, direct curvature calculation gives

$$G_{tt} = 21H^2,$$

$$G_{AB} = -(6\dot{H} + 21H^2)g_{AB}.$$

Therefore the two independent homogeneous Einstein equations are

$$21H^2 = \kappa_8 \rho,$$

$$6\dot{H} + 21H^2 = -\kappa_8 p.$$

Choose dimensionless normalization

$$\kappa_8 = 21.$$

Then

$$H^2 = \rho,$$

$$\dot{H} = -\frac{7}{2}(\rho + p).$$

The acceleration equation follows:

$$\frac{\ddot{a}}{a} = H^2 + \dot{H} = -\frac{5}{2}\rho - \frac{7}{2}p.$$

For a single component with $p=w$ rho, expansion accelerates when

$$w < -\frac{5}{7}.$$

The threshold differs from the familiar four-dimensional value $-1/3$ because seven directions share the scale factor.

8 Chosen nonlinear spinor potential

Select

$$V(S) = \frac{1}{20}S + \frac{19}{20}S^{1/5}.$$

The first term is mass-like. The second is a nonlinear self-interaction. They are both generated by the same rank-16 spinor field; no extra matter fluid is added. Because $\mathbb{C}$ has split signature while the fractional power is evaluated as a real principal power, this model is defined on the positive-condensate cone $S > 0$. The selected initial data have $S=1$, and the exact identity $S=a^{-7}$ preserves that domain for every positive scale factor.

The derivative is

$$V_S = \frac{1}{20} + \frac{19}{100}S^{-4/5}.$$

Energy density and pressure are

$$\rho = \frac{1}{20}S + \frac{19}{20}S^{1/5},$$

$$p = -\frac{19}{25}S^{1/5}.$$

Since $S=a^{-7}$ for the selected normalization,

$$\rho(a) = \frac{1}{20}a^{-7} + \frac{19}{20}a^{-7/5},$$

$$p(a) = -\frac{19}{25}a^{-7/5}.$$

The first contribution has $w=0$ and scales as a^{-7} , the dust law in seven scaled dimensions. The second has

$$w = \frac{1}{5} - 1 = -\frac{4}{5}$$

and scales as $a^{-7/5}$. Since $-4/5 < -5/7$, it can drive acceleration in this model.

9 Complete monomial connection map

For a general positive monomial spinor interaction

$$V_q(S) = \lambda_q S^q,$$

the homogeneous formulas give

$$\rho_q = \lambda_q S^q, \quad p_q = (q - 1)\rho_q, \quad w_q = q - 1,$$

and

$$\rho_q \propto a^{-7q}.$$

This classifies every component in the monomial family:

- $q=1$ gives $w=0$ and ρ proportional to a^{-7} : a pressureless dust-like component.
- $q=8/7$ gives $w=1/7$ and ρ proportional to a^{-8} : a radiation-like component in seven scaled dimensions.
- $q=2/7$ gives $w=-5/7$ and ρ proportional to a^{-2} : the acceleration boundary.
- $0 < q < 2/7$ gives $-1 < w < -5/7$ and dilution slower than a^{-2} : a nonphantom accelerating component.
- $q=0$ gives $w=-1$ and constant density: a cosmological-constant-like limiting term.
- $q < 0$ gives $w < -1$ and density that grows during expansion: phantom-like behavior singular as S approaches zero.

The selected model uses $q=1$ and $q=1/5$. It intentionally contains no $q=0$ constant term, so it does not hide a cosmological constant inside the potential.

For a sum

$$V(S) = \sum_q \lambda_q S^q,$$

the total equation of state is the density-weighted mean

$$w(S) = \frac{\sum_q (q - 1) \lambda_q S^q}{\sum_q \lambda_q S^q}.$$

This is the complete relation between condensate monomials and homogeneous fluid analogies within the stated potential family. It is not a classification of every possible nonlocal, derivative, anisotropic, quantum, or multi-spinor model; that unrestricted set is infinite.

10 Acceleration transition

For the selected two-term potential,

$$\frac{\ddot{a}}{a} = -\frac{1}{8}S + \frac{57}{200}S^{1/5}.$$

The exact transition condition is

$$S_*^{4/5} = \frac{57}{25}.$$

Since $S=a^{-7}$,

$$a_* = \left(\frac{25}{57}\right)^{5/28} = 0.8631436165767085\dots$$

The reported transition time is obtained by linear interpolation in $\mathbf{a}$ between neighboring output samples separated by $\Delta \mathbf{t}=0.01$:

$$t_* = -0.138005293303245\dots$$

The scale factor is analytic; the time carries output-grid interpolation error and should not be interpreted as having 17 digits of physical accuracy.

11 Numerical initial-value problem

The real state has 18 components:

$$y = (a, H, \psi_0, \dots, \psi_{15}).$$

The ODE system is

$$\dot{a} = aH,$$

$$\dot{H} = -\frac{7}{2}(\rho + p),$$

$$\dot{\psi} = -\frac{7}{2}H\psi - V_S\gamma^4\psi.$$

Initial data at $\mathbf{t}=0$ are

$$a(0) = 1, \quad H(0) = 1,$$

$$\psi_0(0) = 1, \quad \psi_{15}(0) = \frac{1}{2},$$

with all other spinor components zero. The exact matrix $\mathbf{C}$ pairs components 0 and 15 symmetrically, so

$$S(0) = 1.$$

Then

$$\rho(0) = 1, \quad H(0)^2 = \rho(0),$$

and the Friedmann constraint is satisfied initially.

CVODE integrates two branches from the same initial state:

```
backward interval = [0,-0.2]
forward interval  = [0, 1.5]
output step       = 0.01
relative tolerance = 1e-11
absolute tolerance = 1e-13 for every component
maximum step      = 0.002
method            = BDF
nonlinear method  = Newton
linear solver      = dense
```

The branches are joined with the initial sample included once, producing 171 ordered rows.

12 Canonical numerical results

The canonical release run reports

```
sample_count=171
solver_steps=1372
rhs_evaluations=1486
acceleration_transition_time=-0.138005293303245
acceleration_transition_scale_factor=0.8631436165767085
maximum_condensate_relative_error=7.441921424927986e-9
maximum_density_relative_error=1.4894832079072834e-9
maximum_friedmann_relative_error=1.203543250505252e-9
maximum_component_ODE_RMS=4.705850657056059e-4
maximum_spatial_Einstein_finite_difference_error=5.702272371471661e-7
```

The finite-difference ODE residual is computed from the 0.01 output grid, not from CVODE's internal smaller steps. A five-point stencil differentiates all 18 emitted state columns. Its truncation error is therefore much larger than the analytic-identity errors, but it distinguishes the implemented spinor rotation sign from the opposite sign by orders of magnitude.

A separate convergence run uses relative tolerance $1\text{e-}12$, absolute tolerance $1\text{e-}14$, and maximum internal step 0.001. It takes 2,294 steps and 2,422 right-hand-side evaluations. Across all 171 output times and all 18 state components, its maximum scaled difference from the canonical run is $6.310776406656671\text{e-}10$. Its maximum condensate, density, and Friedmann errors are respectively $1.6415774988196702\text{e-}9$, $3.2855796418252515\text{e-}10$, and $3.31034411227924\text{e-}10$, each smaller than in the canonical run.

Canonical hashes are

```
geometry.json=6b5eab6b001d69c5face7b179af8a27e3cfe254cf80bc740cdcb7a2855fcdf67
wolfram-report.json=f81a902f7efc2a36bef1bacdbc4682a278f4907383970a8383f565efd602dc38
generated.rs=fa3aba370c1039f17b83fe86a65d75e45be6448aa7bd025dc0db6d0e79e61f60
history.csv=9553b36f201ef43a92c7de3fe2f46457a592e55285e6220d8aa7af6e36a26f9c
summary.json=6b15d084178d01aaed37b84b4bcf8f5246a6941bbe8366ad9c0c3db099eea0f1
```

At the earliest sample $t=-0.2$:

```
a=0.8018695
H=1.2364004
w=-0.6772517
dust_like_fraction=0.1534354
negative_pressure_fraction=0.8465646
acceleration=-0.1981471
```

At the normalized present sample $t=0$:

```
a=1
H=1
w=-0.76
dust_like_fraction=0.05
negative_pressure_fraction=0.95
acceleration=0.16
```

At the latest sample $t=1.5$:

```
H=0.4808740
w=-0.7998523
dust_like_fraction=0.0001846
negative_pressure_fraction=0.9998154
acceleration=0.0692524
```

The two fractions sum to one by construction. They describe contributions to the effective homogeneous spinor energy density, not independently detected particle populations.

13 Independent verification

The release does not accept the reduced equations or the solver's stored summary as proof. The exact Python model checker performs four rational checks:

- reconstruction of the initial positive condensate from the actual $\mathbf{C}$ and spinor components;
- direct evaluation of the on-shell Lagrangian and every component of the corrected action stress tensor using the exact 16 by 16 gamma matrices;
- reconstruction of the full Einstein tensor from second-order metric jets at two independent rational choices of $\mathbf{a}$, $\mathbf{H}$, and $\mathbf{Hdot}$; and
- all 64 components of $\mathbf{G}_{\mu \nu} = 21 \mathbf{T}_{\mu \nu}$ at the normalized initial state, including vanishing off-diagonal components.

The trajectory checker reloads `history.csv` and performs 22 base checks:

- file, schema, signature, state-dimension, source-hash, and parameter checks;
- exact output headers, sample count, time grid, finite values, and initial data;
- recomputation of $\mathbf{S}$, ρ , $\mathbf{p}$, $\mathbf{w}$, both component fractions, and acceleration from all 16 spinor columns;
- positivity of $\mathbf{a}$ and $\mathbf{S}$ throughout the implemented fractional-power domain;
- analytic comparisons with $\mathbf{S} = \mathbf{a}^{-7}$ and the corresponding density;
- direct Friedmann residual $\mathbf{H}^2 - \rho$;
- five-point finite differences for $\mathbf{a}$, $\mathbf{H}$, and all 16 spinor components;
- an independently differentiated spatial Einstein residual $6 \mathbf{Hdot} + 21 \mathbf{H}^2 + 21 \mathbf{p}$;
- exact acceleration-transition scale and interpolated transition time;
- canonical SHA-256 values, solver statistics, and verdict.

The release command supplies both optional evidence sets, increasing that total to 24: byte-identical replay of the canonical run and componentwise comparison with the tighter convergence run. If either path is omitted, its check is reported as not performed rather than vacuously true.

The Rust crate also contains four tests for exact matrix adjoint properties, the present Einstein constraints, condensate derivative, rejection of nonpositive scale factors or condensates, CVODE smoke integration, analytic errors, and acceleration.

14 What may be connected to dark matter

The mass-like term

$$\rho_{\text{dust}} = \frac{1}{20} S$$

has zero pressure and dilutes as $\mathbf{a}^{-7}$, exactly like homogeneous pressureless matter in seven scaled dimensions. This is a precise background-level connection to a dark-matter-like fluid.

It is not enough to call the component dark matter physically. No galactic clustering, perturbation growth, sound speed, lensing, halo structure, standard-model coupling, relic abundance, or compactification to four dimensions is calculated. The label "dust-like" is therefore used in emitted artifacts.

15 What may be connected to dark energy

The nonlinear term

$$\rho_{\text{neg}} = \frac{19}{20} S^{1/5}$$

has $w=-4/5$, satisfies the 8D acceleration condition, and eventually dominates because it dilutes more slowly than the mass term. This is a precise background-level connection to a nonphantom dark-energy-like component.

It is not a cosmological constant: its density changes with $\mathbf{a}$, and the selected action contains no constant term. It is not scalar quintessence: the dynamical section is a rank-16 real spinor, and its covariant derivative contains the Spin connection.

It is also not an observational dark-energy result. The base has split signature, there is no compactification map to observed $(3,1)$ spacetime, and no likelihood analysis is performed.

16 Other possible connections and excluded claims

Within local polynomial or monomial potentials, changing powers and coefficients can produce mixtures analogous to dust, radiation, accelerating nonphantom energy, a constant term, or phantom behavior. Derivative couplings, curvature couplings such as $R \mathbf{S}$, torsion, nonmetricity, multiple spinors, and anisotropic condensates would produce additional possibilities, but none is silently added here.

The present result does not establish:

1. a physical theory with four observed time directions;
2. a ghost-free or hyperbolic perturbation system;
3. stability against inhomogeneous spinor or metric perturbations;
4. dimensional reduction to $(3,1)$ spacetime;
5. a quantum theory of a Grassmann-odd fermion;
6. dark-matter clustering or a dark-energy observational fit;
7. uniqueness of the potential or initial data;
8. a particle-generation mechanism;

9. compatibility with standard energy conditions;
10. microscopic derivation from split octonions.

These are separate research questions. The numerical solution establishes a well-specified classical homogeneous background and nothing broader.

17 Complete Windows commands

Open PowerShell and run every command below:

```
Set-Location C:\Users\nsh\Developer\code\vscode\dirac
git submodule update --init --recursive
```

```
python scripts\backup_files.py '
    studies\einstein_spinor_44\src\generated.rs '
    artifacts\einstein-spinor-44\history.csv '
    artifacts\einstein-spinor-44\summary.json '
    provenance\EINSTEIN_SPINOR_44.tex '
    provenance\EINSTEIN_SPINOR_44.pdf
```

```
New-Item -ItemType Directory -Force -Path '
    build\phase5, '
    build\phase5\einstein-spinor-repeat, '
    build\phase5\einstein-spinor-refined, '
    build\phase5\gravity-pdf-a, '
    build\phase5\gravity-pdf-b | Out-Null
```

```
python scripts\build_curved_spin_geometry.py
wolframscript -file scripts\verify_curved_spin_geometry.wls -- '
    artifacts\curved-spin-geometry\wolfram-report.json
python scripts\check_curved_spin_geometry.py
python scripts\check_einstein_spinor_model.py
```

```
python scripts\generate_einstein_spinor_constants.py
python scripts\generate_einstein_spinor_constants.py '
    --output build\phase5\einstein-spinor-generated-repeat.rs
```

```
cargo fmt -p einstein_spinor_44 -- --check
cargo clippy -p einstein_spinor_44 --all-targets -- -D warnings
cargo test -p einstein_spinor_44
python -m unittest discover -s tests -v
```

```
cargo run --release -p einstein_spinor_44 -- '
    --output artifacts\einstein-spinor-44
cargo run --release -p einstein_spinor_44 -- '
    --output build\phase5\einstein-spinor-repeat
cargo run --release -p einstein_spinor_44 -- '
    --output build\phase5\einstein-spinor-refined '
    --relative-tolerance 1e-12 '
    --absolute-tolerance 1e-14 '
```

```

--maximum-step 0.001

python scripts\check_einstein_spinor_44.py '
--repeat build\phase5\einstein-spinor-repeat '
--refined build\phase5\einstein-spinor-refined

python scripts\build_dissertation_tex.py --strip-heading-numbers '
--input provenance\EINSTEIN_SPINOR_44.md '
--output provenance\EINSTEIN_SPINOR_44.tex
python scripts\build_dissertation_tex.py --strip-heading-numbers '
--input provenance\EINSTEIN_SPINOR_44.md '
--output build\phase5\EINSTEIN_SPINOR_44-repeat.tex

pdflatex -interaction=nonstopmode -halt-on-error '
-jobname=EINSTEIN_SPINOR_44 '
-output-directory=build\phase5\gravity-pdf-a '
provenance\EINSTEIN_SPINOR_44.tex
pdflatex -interaction=nonstopmode -halt-on-error '
-jobname=EINSTEIN_SPINOR_44 '
-output-directory=build\phase5\gravity-pdf-a '
provenance\EINSTEIN_SPINOR_44.tex
pdflatex -interaction=nonstopmode -halt-on-error '
-jobname=EINSTEIN_SPINOR_44 '
-output-directory=build\phase5\gravity-pdf-a '
provenance\EINSTEIN_SPINOR_44.tex

pdflatex -interaction=nonstopmode -halt-on-error '
-jobname=EINSTEIN_SPINOR_44 '
-output-directory=build\phase5\gravity-pdf-b '
build\phase5\EINSTEIN_SPINOR_44-repeat.tex
pdflatex -interaction=nonstopmode -halt-on-error '
-jobname=EINSTEIN_SPINOR_44 '
-output-directory=build\phase5\gravity-pdf-b '
build\phase5\EINSTEIN_SPINOR_44-repeat.tex
pdflatex -interaction=nonstopmode -halt-on-error '
-jobname=EINSTEIN_SPINOR_44 '
-output-directory=build\phase5\gravity-pdf-b '
build\phase5\EINSTEIN_SPINOR_44-repeat.tex

python scripts\check_provenance_pdf.py --edition einstein-spinor-44 '
build\phase5\gravity-pdf-a\EINSTEIN_SPINOR_44.pdf '
--repeat build\phase5\gravity-pdf-b\EINSTEIN_SPINOR_44.pdf

$generatedA = (Get-FileHash '
studies\einstein_spinor_44\src\generated.rs '
-Algorithm SHA256).Hash
$generatedB = (Get-FileHash '
build\phase5\einstein-spinor-generated-repeat.rs '
-Algorithm SHA256).Hash
if ($generatedA -ne $generatedB) { throw "Generated constants changed bytes" }
$historyA = (Get-FileHash '
artifacts\einstein-spinor-44\history.csv '
-Algorithm SHA256).Hash
$historyB = (Get-FileHash '

```

```

    build\phase5\einstein-spinor-repeat\history.csv '
    -Algorithm SHA256).Hash
if ($historyA -ne $historyB) { throw "Einstein-spinor CSV changed bytes" }
$summaryA = (Get-FileHash '
    artifacts\einstein-spinor-44\summary.json '
    -Algorithm SHA256).Hash
$summaryB = (Get-FileHash '
    build\phase5\einstein-spinor-repeat\summary.json '
    -Algorithm SHA256).Hash
if ($summaryA -ne $summaryB) { throw "Einstein-spinor summary changed bytes" }

$warningPattern = '^!|LaTeX Warning|Package .* Warning|'
$warningPattern += 'Overfull|Underfull|Undefined control sequence'
$issues = Select-String '
    -Path build\phase5\gravity-pdf-a\EINSTEIN_SPINOR_44.log, '
        build\phase5\gravity-pdf-b\EINSTEIN_SPINOR_44.log '
    -Pattern $warningPattern
if ($issues) { throw "Einstein-spinor LaTeX warnings remain" }

Copy-Item build\phase5\gravity-pdf-a\EINSTEIN_SPINOR_44.pdf '
    provenance\EINSTEIN_SPINOR_44.pdf -Force

.\scripts\verify_phase5_curved_spin_gravity.ps1

```

The final line must be

```
phase5_curved_spin_gravity_verification=OK
```

18 Complete Git Bash or WSL commands

Run every command below:

```

if [[ -d /c/Users/nsh/Developer/code/vscode/dirac ]]; then
    cd /c/Users/nsh/Developer/code/vscode/dirac
else
    cd /mnt/c/Users/nsh/Developer/code/vscode/dirac
fi
git submodule update --init --recursive
source ./scripts/resolve_wolframscrip.sh
python_command="$(resolve_windows_command python.exe)"
cargo_command="$(resolve_windows_command cargo.exe)"
pdflatex_command="$(resolve_miktex_pdflatex)"

"$python_command" scripts/backup_files.py \
    studies/einstein_spinor_44/src/generated.rs \
    artifacts/einstein-spinor-44/history.csv \
    artifacts/einstein-spinor-44/summary.json \
    provenance/EINSTEIN_SPINOR_44.tex \
    provenance/EINSTEIN_SPINOR_44.pdf

```

```

rm -rf build/phase5/einstein-spinor-repeat \
    build/phase5/einstein-spinor-refined \
    build/phase5/gravity-pdf-a build/phase5/gravity-pdf-b
mkdir -p build/phase5/einstein-spinor-repeat \
    build/phase5/einstein-spinor-refined \
    build/phase5/gravity-pdf-a build/phase5/gravity-pdf-b

"$python_command" scripts/build_curved_spin_geometry.py
wolframscript_command="$(resolve_wolframscript)"
"$wolframscript_command" -file scripts/verify_curved_spin_geometry.wls -- \
    artifacts/curved-spin-geometry/wolfram-report.json
"$python_command" scripts/check_curved_spin_geometry.py
"$python_command" scripts/check_einstein_spinor_model.py

"$python_command" scripts/generate_einstein_spinor_constants.py
"$python_command" scripts/generate_einstein_spinor_constants.py \
    --output build/phase5/einstein-spinor-generated-repeat.rs

"$cargo_command" fmt -p einstein_spinor_44 -- --check
"$cargo_command" clippy -p einstein_spinor_44 --all-targets -- -D warnings
"$cargo_command" test -p einstein_spinor_44
"$python_command" -m unittest discover -s tests -v

"$cargo_command" run --release -p einstein_spinor_44 -- \
    --output artifacts/einstein-spinor-44
"$cargo_command" run --release -p einstein_spinor_44 -- \
    --output build/phase5/einstein-spinor-repeat
"$cargo_command" run --release -p einstein_spinor_44 -- \
    --output build/phase5/einstein-spinor-refined \
    --relative-tolerance 1e-12 \
    --absolute-tolerance 1e-14 \
    --maximum-step 0.001
"$python_command" scripts/check_einstein_spinor_44.py \
    --repeat build/phase5/einstein-spinor-repeat \
    --refined build/phase5/einstein-spinor-refined

"$python_command" scripts/build_dissertation_tex.py \
    --strip-heading-numbers \
    --input provenance/EINSTEIN_SPINOR_44.md \
    --output provenance/EINSTEIN_SPINOR_44.tex
"$python_command" scripts/build_dissertation_tex.py \
    --strip-heading-numbers \
    --input provenance/EINSTEIN_SPINOR_44.md \
    --output build/phase5/EINSTEIN_SPINOR_44-repeat.tex

for pass in 1 2 3; do
    "$pdflatex_command" -interaction=nonstopmode -halt-on-error \
        -jobname=EINSTEIN_SPINOR_44 \
        -output-directory=build/phase5/gravity-pdf-a \
        provenance/EINSTEIN_SPINOR_44.tex
    "$pdflatex_command" -interaction=nonstopmode -halt-on-error \
        -jobname=EINSTEIN_SPINOR_44 \
        -output-directory=build/phase5/gravity-pdf-b \
        build/phase5/EINSTEIN_SPINOR_44-repeat.tex

```

done

```
"$python_command" scripts/check_provenance_pdf.py \
  --edition einstein-spinor-44 \
  build/phase5/gravity-pdf-a/EINSTEIN_SPINOR_44.pdf \
  --repeat build/phase5/gravity-pdf-b/EINSTEIN_SPINOR_44.pdf

generated_a="$(sha256sum \
  studies/einstein_spinor_44/src/generated.rs | cut -d' ' -f1)"
generated_b="$(sha256sum \
  build/phase5/einstein-spinor-generated-repeat.rs | cut -d' ' -f1)"
test "$generated_a" = "$generated_b"
history_a="$(sha256sum \
  artifacts/einstein-spinor-44/history.csv | cut -d' ' -f1)"
history_b="$(sha256sum \
  build/phase5/einstein-spinor-repeat/history.csv | cut -d' ' -f1)"
test "$history_a" = "$history_b"
summary_a="$(sha256sum \
  artifacts/einstein-spinor-44/summary.json | cut -d' ' -f1)"
summary_b="$(sha256sum \
  build/phase5/einstein-spinor-repeat/summary.json | cut -d' ' -f1)"
test "$summary_a" = "$summary_b"
warning_pattern='^!|LaTeX Warning|Package .* Warning|'
warning_pattern+='Overfull|Underfull|Undefined control sequence'
! grep -Ei "$warning_pattern" \
  build/phase5/gravity-pdf-a/EINSTEIN_SPINOR_44.log \
  build/phase5/gravity-pdf-b/EINSTEIN_SPINOR_44.log
cp build/phase5/gravity-pdf-a/EINSTEIN_SPINOR_44.pdf \
  provenance/EINSTEIN_SPINOR_44.pdf

bash ./scripts/verify_phase5_curved_spin_gravity.sh
```

19 Files and ownership boundaries

```
studies/einstein_spinor_44/Cargo.toml
studies/einstein_spinor_44/src/lib.rs
studies/einstein_spinor_44/src/main.rs
studies/einstein_spinor_44/src/generated.rs
scripts/generate_einstein_spinor_constants.py
scripts/check_einstein_spinor_model.py
scripts/check_einstein_spinor_44.py
artifacts/einstein-spinor-44/history.csv
artifacts/einstein-spinor-44/summary.json
provenance/EINSTEIN_SPINOR_44.md
provenance/EINSTEIN_SPINOR_44.tex
provenance/EINSTEIN_SPINOR_44.pdf
```

The solver submodule is read-only. The new equations, generated matrices, output format, checks, and documentation belong to the application repository.

20 Conclusion

A new signature (4,4) Einstein system has been defined without a scalar quintessence field and without a cosmological constant. Its only source is a commuting classical real rank-16 $\text{Spin}(4,4)$ spinor with a two-term invariant potential. The mass-like term is exactly dust-like at the homogeneous level; the nonlinear term has $w=-4/5$ and produces acceleration after the exact scale $a_*= (25/57)^{(5/28)}$.

The solution is approximate numerically but tightly checked: analytic condensate and density laws, both Einstein equations, all 18 ODE components, source hashes, solver statistics, canonical output hashes, and deterministic replay all pass. The result is a reproducible split-signature background model, not an observational identification of dark matter or dark energy.